

No. of Functional Features Included in the Solution

Date	15 March 2024
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Dataset Loading	Loads HDI dataset from CSV file	Data Module	Done	High
2	Data Preprocessing	Cleans missing values, removes duplicates, and scales features	Preprocessing Module	Done	High
3	Data Visualization	Generates heatmaps, pair plots, histograms, and correlation graphs	EDA Module	Done	Medium
4	Machine Learning Model Training	Trains Logistic Regression, Decision Tree, and Random Forest models	ML Module	Done	High
5	Model Evaluation	Calculates Accuracy, Confusion Matrix, Precision, Recall, and F1-Score	Evaluation Module	Done	High
6	HDI Category Prediction	Predicts HDI category (Low, Medium, High, Very High) based on user inputs	Prediction Module	Done	High
7	Flask Web Application	Provides a web interface for users to enter inputs and view prediction results	Frontend & Backend	Done	High

8	Model Saving & Loading	Saves the trained model using Pickle and loads it during prediction	Deployment Module	Done	Medium
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Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6

Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	HTML Forms, Prediction Result Page	Input form, Predict button, Result display
2	Backend / Logic	Flask Backend, Prediction Logic	Input validation, Model loading, Prediction
3	Database / Storage	Dataset & Pickle Model Storage	CSV dataset, model.pkl
4	API / Integration	Flask Routing	/ Home page, /predict Prediction endpoint
5	Security / Authentication	Basic Input Validation	Numeric input validation, Exception handling